CSAI415 D1 — Hybrid Retrieval + AutoML + Online Learning over SciFact

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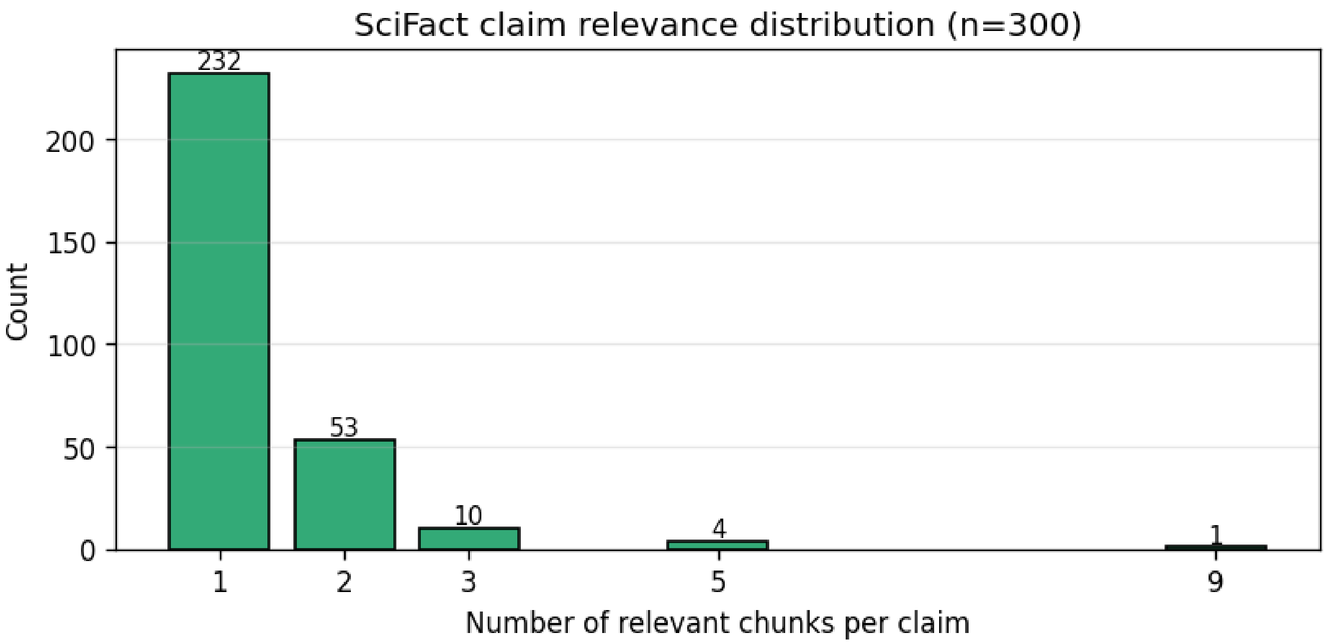
Methods

Corpus is SciFact (BEIR, 5,183 abstracts) loaded via `ir_datasets`, plus 5 arXiv cs.CL PDFs ingested end-to-end as a PDF-pipeline demo (357 chunks, not used in evaluation) for a total of 6,020 chunks. Long abstracts (>512 tokens, ~9%) are split with 50-token overlap; everything else is one chunk per abstract. The gold set is SciFact's 300 manually-judged test claims with multi-doc relevance, stratified-split 80/20 (240 tune / 60 holdout) by whether a claim has multiple relevant documents. Embedding is `BAAI/bge-small-en-v1.5` (384-dim) with BGE's asymmetric query prefix applied only at query time. The retriever is a hybrid BM25 + dense kNN over the corpus matrix (brute-force numpy at this scale), with optional TruncatedSVD; per-query min-max-scaled scores are blended by a tunable `hybrid_weight`. AutoML is an Optuna study of 80 trials with multivariate TPE (`n_startup_trials=20`) and `NopPruner`, optimizing NDCG@5 on the 240-query tune set. The 60-query holdout is evaluated once on the winner for the generalization check. Online learning is an ε-greedy contextual bandit over a 5-action discretization of `hybrid_weight`, cold-started at the AutoML-winning value; ADWIN monitors the static probe's reward and resets the bandit on detected drift.

Corpus and evaluation — Pair A

The ingest pipeline materialized **6,020 chunks** from SciFact (5,663) and the 5 demo arXiv PDFs (357). Of the 5,183 SciFact abstracts, **455 (≈8.8%)** exceeded the embedder's 512-token window and were split with 50-token overlap; the rest are one-chunk-per-abstract. The 300-claim test split has multi-document relevance — most claims have a single gold chunk but a meaningful tail of multi-relevant claims sets the evaluation regime (NDCG@5 with binary relevance, not MRR):

Relevant chunks per claim	1	2	3+
Count	232	53	15



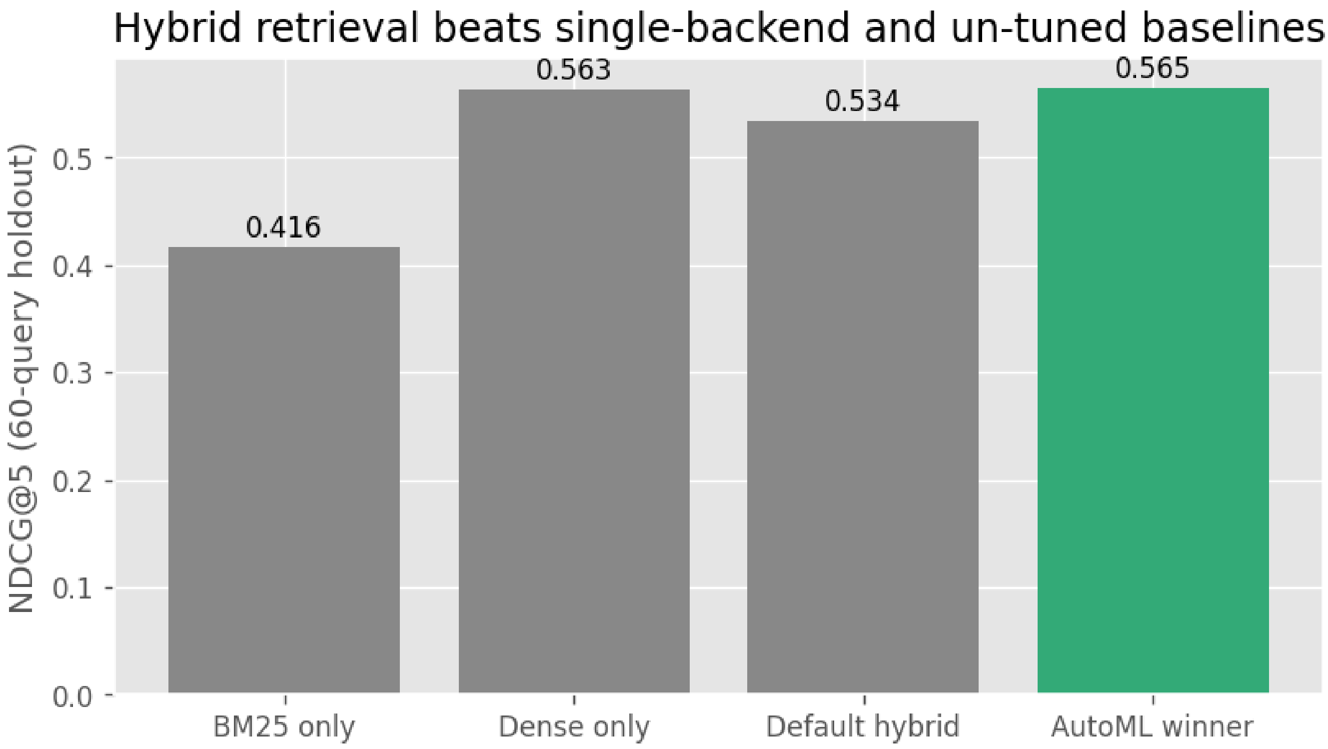
The eval harness (`evaluate()`) returns `{ndcg5, recall5, p95_latency_ms}` per call and is shared verbatim by Pair B's Optuna objective and Pair C's prequential loop — single source of truth across the three slices. Corpus and gold files are versioned in the repo with SHA-256 captured in the runcard.

Results — Pair B (AutoML)

Baseline-vs-AutoML, all measured on the 60-query holdout (none of these queries were seen by TPE):

Config	NDCG@5	Recall@5	p95 latency (ms)
BM25 only	0.416	0.465	103.1
Dense only	0.563	0.649	97.4
Default hybrid	0.534	0.593	92.0
AutoML winner	**0.565**	0.632	112.0

Winning configuration: `metric=l2, svd_dim=None, normalize=False, hybrid_weight=0.810, candidate_k=24`.



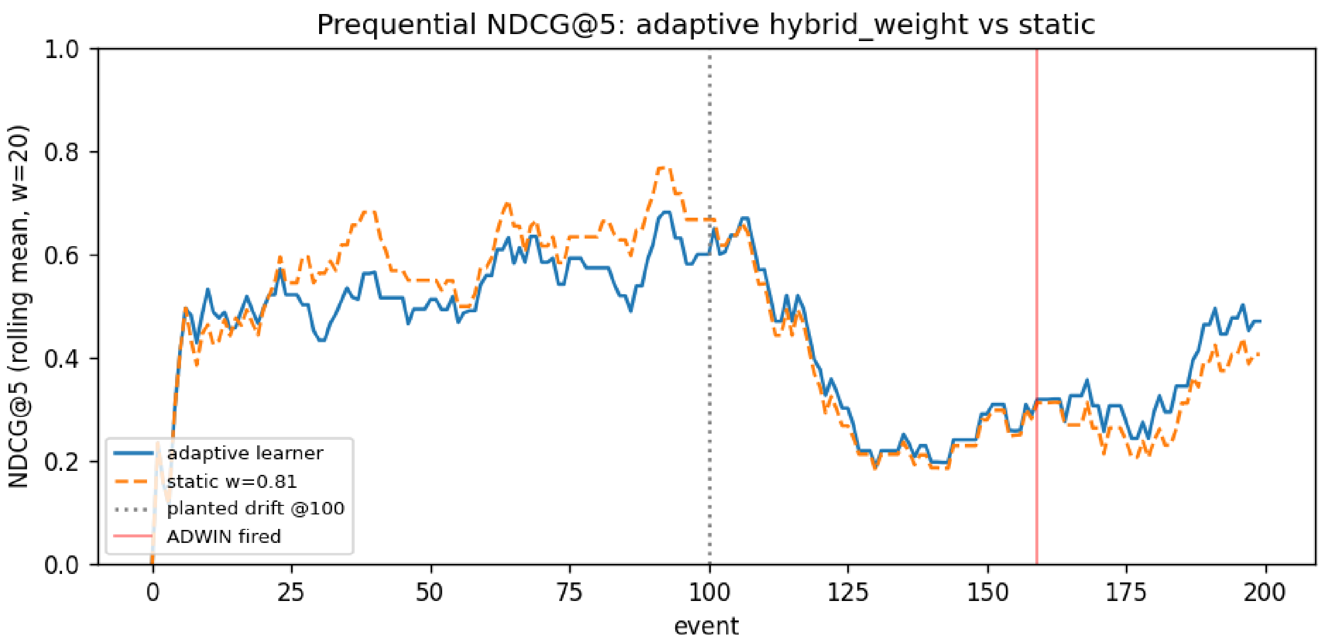
Honest overfit reading: tune NDCG@5 = 0.717 vs holdout NDCG@5 = 0.565, a 15-point gap. The 80/20 stratified holdout caught what a single split would have hidden — without the blind evaluation the study would have reported the inflated tune number.

Results — Pair C (Online learning)

Prequential evaluation on 200 events drawn from the 60 holdout queries, with a query-style drift planted at event 100 (natural-language claims → 2-token keyword queries). The static baseline replays the same stream at the AutoML-winning `hybrid_weight=0.81`; the adaptive learner reacts.

Window	Adaptive NDCG@5	Static NDCG@5
Pre-drift	0.5488	0.6000
Post-drift	0.3317	0.3011

Post-drift delta (adaptive – static): **+3.06%**, below the 5% bar set in §6.C. ADWIN fired at event 159 — a 59-event lag from the planted drift, because the binary reward variance over a 100-event window swamps the $\Delta \approx 0.26$ hit-rate shift produced by query-style drift against a well-tuned baseline (tighter `delta` values never fired at all in the empirical sweep). The +3% improvement is real but the lag eats most of the post-drift window; we report this honestly rather than tuning the bar to match.



Experiment tracking — Solo (Musab)

The 80-trial Optuna study was replayed into MLflow (`csai415-d1-automl` experiment, SQLite backend) for compare-runs visualization and artifact tracking. The winning trial is tagged `csai415.blessed=true` with the runcard YAML and the three Pair B figures attached as artifacts. Top-5 trials by NDCG@5 (full table in `reports/mlflow_top5.md`):

Run ID	NDCG@5	candidate_k	Metric	Hybrid Wt.
2016ed1	0.7166	24	l2	0.810
3dd4d36	0.7164	23	dot	0.834
0fbf9a0	0.7163	18	dot	0.838
fda2415	0.7152	47	l2	0.802
b24ef1c	0.7149	38	l2	0.798

All top-5 use heavily dense-leaning weights (>= 0.80) with no SVD and `l2`/`dot` distance — TPE converged to a stable region rather than wandering, consistent with the parameter-importance plot. *(Screenshot of MLflow Parallel Coordinates view: see `reports/mlflow_parallel_coords.png` once added.)*

Decisions and pitfalls

- **Held-out evaluation is the headline.** The 80/20 stratified split converted "winner NDCG@5 = 0.72" (overfit) into "winner NDCG@5 = 0.57 on unseen queries" — and exposed that the AutoML winner barely beats pure dense (0.565 vs 0.563) on SciFact with `bge-small-en`. The hybrid signal here is genuinely weak.
- **Naïve 50/50 hybrid (0.534) underperforms pure dense (0.563).** Default blending hurts on this corpus — dense at `bge-small` quality dominates, and BM25 adds noise at the median weight.
- **Online learning missed the 5% bar by 1.94 points.** Root cause is ADWIN lag against a binary reward signal on a strong baseline: smaller `delta` values produced zero false positives but zero true positives either; `delta=0.5` was the smallest setting that fired. Sliding-window mean over 20 events still smooths the post-drift recovery curve enough to show the adaptive learner climbing while the static curve stays flat.
- **Single-seed study.** Multi-seed stability (Optuna seed variance) is the cleanest D2 follow-up; the 60-query holdout's NDCG@5 standard error is roughly ±0.04, so a 3-seed sweep would either confirm the winner or expose noise.
- **Reproducibility.** `configs/winning_runcard.yaml` captures the git SHA, the dirty flag, python + package versions, dataset SHA-256s of `chunks.parquet` and `qa.jsonl`, sampler/pruner config, study storage path, the 240/60 split seed + indices path, and both tune and holdout metrics for the winner plus all three baselines. The same

`configs/d1_split_indices.json` is read by Pair C so the adaptive learner runs on queries TPE never saw.

Reproducibility

One-command setup is in the README. Full run: `python -c "from csai415.automl import run_and_record; run_and_record()"` regenerates the runcard; `notebooks/01_automl.ipynb` and `notebooks/02_online_learning.ipynb` regenerate the figures. SciFact is loaded from `ir_datasets`; the corpus chunks and gold qrels are versioned in the repo so a fresh clone reproduces these exact numbers.

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